W-T-215C
August 25, 1969
SUPERSEDING
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FEDERAL SPECIFICATION

TEST SET; GENERATOR AND VOLTAGE REGULATOR, AUTOMOTIVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of test set suitable for making "on the vehicle" tests of automotive low-voltage electrical systems on passenger and commercial type vehicles.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

FSC 4910

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-W-80	- Window, Observation, Acrylic Base, Anti-electrostatic, Transparent (for Indicating Instrument).
MIL-T-152	- Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-G-3787	- Glass Laminated, Flat; (Except Aircraft).
MIL-E-17555	- Electronic and Electrical Equipment Accessories, and Repair Parts; Packaging and Packing of.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

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MIL-STD-130 - Identification Marking of US Military Property.

MIL-STD-681 - Identification Coding and Application of Hookup Wire and Lead Wire.

(Copies of Military Specifications, Standards, Drawings, and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS), Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

National Electrical Manufacturers' Association (NEMA):

Standards

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, New York 10017.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one complete test set for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

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3.2 Design. The test set shall be new and of such design as to permit on the vehicle testing of nonwaterproof 6, 12, and 24-volt electrical systems of passenger and commercial type vehicles. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair.

3.3 Materials. Materials shall be as specified herein. Materials not specifically designated shall be of good commercial quality. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium, shall be avoided wherever possible.

3.4 Construction. The test set shall be complete so that when placed in use it shall be capable of performing the operations specified herein.

3.4.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of the test set.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be performed only where these operations are required in the original design.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used the conductors and terminal parts shall be mechanically secured before soldering. Only resin base fluxes shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4.5 Threads. All threads shall be in the inch system and comply with NBS H28.

3.4.6 Safety devices. Each test set shall incorporate safety devices, if required, to prevent conditions hazardous to the operator. Protection against electric shock shall be provided. When specified (see 6.2), special safety devices shall be provided as required by the procuring activity.

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3.4.7 Interchangeability. All replaceable parts shall be manufactured to definite tolerances and clearances in order that any such part of a particular type or model may be replaced or adjusted without requiring modification.

3.4.8 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smoothed and all edges shall be rounded or beveled.

3.4.9 Protective finishes. Exposed metal and working surfaces, except when parts are inherently corrosion resistant, shall be finished to resist corrosion. Exposed metal is defined as metal exposed to atmospheric conditions specified herein.

3.5 Performance and product characteristics.

3.5.1 Initial accuracy. With the test set in its normal operating position and in an ambient air temperature of 80° ± 10°F., the initial accuracy of the meters, including internal circuits and test leads, shall be within 2 percent of end-scale value for the voltmeter and within 3 percent of the full-scale value for the ammeter.

3.5.2 Accuracy after tests. After completion of all of the tests for requirements specified in 3.5.3 through 3.5.8 (excluding 3.5.3.2.1 and 3.5.3.2.2), and after the test set is stabilized at an ambient air temperature of 80° ± 10°F., the overall accuracy shall be within 1 percent of full-scale value from the reading taken as specified in 3.5.1.

3.5.3 Temperature and humidity conditions.

3.5.3.1 Storage temperatures. The test set and all components thereof shall be able to withstand storage without damage in a temperature of minus 65°F. for periods up to 72 hours, and a 24-hour cycle of temperature changes starting with 10 hours, at plus 90°F., followed by 5 hours of steady increase to plus 155°F., 4 hours at plus 155°F., and 5 hours of steady decrease to plus 90°F. Following these extreme exposures and when stabilized at plus 80°F., the test set shall conform to the accuracy requirements of 3.5.2.

3.5.3.2 Operating temperature. Meters shall indicate freely at all parts of the scale when the test set is operated in an ambient air temperature of minus 25° and + 115°F.

3.5.3.2.1 Low temperature effect. The low temperature effect on the test set, when operated in an ambient air temperature of minus 25°F., shall not change the accuracy more than 2 percent of full-scale value

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from the readings obtained at plus 80°F., preliminary to low temperature operations.

3.5.3.2.2 High temperature effect. The high temperature effect on the test set, when operated in an ambient air temperature of plus 115°F., shall not change the accuracy more than 1 percent of full-scale value from the readings obtained at plus 80°F., preliminary to high temperature operations.

3.5.3.3 Humidity conditions. The test set shall operate within the accuracy requirements of 3.5.2 after exposure to 20 hours of relative humidity between 93 and 97 percent at air temperatures from plus 80° + 85°F., followed by 4 hours of 100 percent relative humidity with condensation at plus 75° + 80°F.; air movement and radiation negligible.

3.5.4 Momentary overload. The meters of the test set shall be capable of withstanding a series of momentary overloads as specified herein. Upon completion, except for setting of the pointer to zero using the external adjustment screw, no additional repairs or adjustment shall be required.

3.5.4.1 Ammeter circuit. The ammeter and its associated circuit shall be capable of withstanding, without damage, a momentary overload of three times full-scale value for one second "on" and 60 seconds "off" for a total of five cycles, followed by a 5 second "on" period. At the end of that time, the meter shall be capable of operating within the accuracy requirements of 3.5.2.

3.5.4.2 Voltmeter circuit. The voltmeter and its associated circuit shall be capable of withstanding, without damage, a momentary overload of 2.8 times the end-scale value for the periods specified in 3.5.4.1. At the end of that time, the meter shall be capable of operating within the accuracy requirements of 3.5.2.

3.5.5 Sustained overload. The meters of the test set shall be capable of withstanding an overload of 20 percent greater than end-scale values, sustained for a period of 8 hours. At the end of that time, no damage shall have occurred and the meters shall be capable of operating within the accuracy requirements of 3.5.2.

3.5.6 Vibration. The test set shall be capable of withstanding, without damage, a simple harmonic motion having an amplitude of 1/32 inch (1/16 inch maximum total excursion), the frequency being varied uniformly between the approximate limits of 5 and 55 cycles per second (c.p.s.). The entire frequency range, from 5 to 55 c.p.s. and return to 5 c.p.s., shall be traversed in 1 minute. At the end of this test, no damage shall have occurred, and the test set shall be capable of operating within the accuracies specified in 3.5.2.

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3.5.7 Shock. The test set shall be capable of withstanding, without damage, the shock resulting from being dropped a vertical distance of 8 inches along each of its three major axes onto a concrete surface. At the end of this test, no damage shall have occurred, and the test set shall be capable of operating within the accuracies specified in 3.5.2.

3.5.8 Dielectric strength. The test set shall be capable of withstanding, without damage, for 1 minute, 500 volts root mean square (r.m.s.) applied between all insulated circuits and between insulated circuits and external metal parts. At the conclusion, the test set shall be capable of operating within the accuracies specified in 3.5.2.

3.5.9 Contact life. Rheostat contacts and all switches furnished shall be capable of being cycled continuously through all contacts at a rate of 10 to 16 cycles per minute (c.p.m.) for a total of 5,000 cycles each. The switches shall be capable of carrying a current of 150 percent of the maximum normal current in its electrical circuit during the closed contact position throughout this test. The voltage drop across any switch or contact shall not exceed 50 millivolts when the switch or contact is in the closed position. The rheostat shall not show an open circuit at any position other than at the "open" position while its rated current is applied throughout this test.

3.5.10 Dimensions. Unless otherwise specified (see 6.2), the overall dimensions of the test set shall not exceed 8 by 14 by 10 inches, or the equivalent volume in a practical shape.

3.6 Detail of components. The test set shall consist of the following principal components.

3.6.1 Enclosure. The enclosure of the test set shall be capable of meeting the requirements and satisfactorily passing the tests specified herein. The enclosure shall be provided with facility for heat dissipation during testing regardless of test set position. A carrying handle shall be provided on the enclosure and shall be of sufficient size to allow for a comfortable hand grip. Doors or covers shall be provided with piano-type hinges. Fixed joints shall be overlapped and welded throughout. Rubber rest buttons shall be provided at each corner of the base of the test set, and at each corner of the side opposite the carrying handle. Storage space shall be provided in the enclosure for the handbook of instructions, leads, and required ammeter shunts.

3.6.1.1 Instrument panel. The instrument panel shall provide for mounting of all electrical component parts either on the panel itself or on a chassis attached to the panel, so that when the panel is removed from the enclosure, all component parts are removed from the enclosure.

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3.6.2 Meters. One voltmeter and one ammeter designed for panel mounting shall be provided. The meters shall be flush mounted with resilient shock cushion on the meter panel. The meter movements shall have jeweled bearings, aluminum-nickel-iron cobalt alloy or equal permanent magnets, and D'Arsonval moving coil construction.

3.6.2.1 Meter housing. The meter housing shall be constructed of molded thermosetting plastic material or metal. Plastic, if used, shall conform to MIL-M-14.

3.6.2.2 Pointers. Pointers shall be of the knife-edge type and of rigid construction. When the meters are not provided with means for avoiding errors in reading due to parallax, the clearance between the pointer index and the graduated scale shall not exceed 0.10 inch or be less than 0.03 inch. The pointers shall contrast in color to the dial background.

3.6.2.3 Windows. Unless otherwise specified (see 6.2), the meters shall be provided with windows of glass or plastic material, free from discoloration, scratches, striae, electrostatic effect, or other detrimental defects which may interfere in any way with normal reading of the meters. Glass windows, when used, shall be in accordance with class II, laminated glass, specified in MIL-G-3787. Plastic windows, when used, shall be in accordance with MIL-W-80, shall be a minimum of 3/32 inch thick, and shall be treated on both sides with an antistatic solution, silicon base preferred. The windows shall be secured by clamps or by other suitable means. Cement alone is not considered sufficient for holding the windows in place.

3.6.2.4 Dials. The meter dials shall be made of nonferrous metal and shall have bonded dull-white background with contrasting numerals and letters printed therein. The mean-scale arc length shall be not less than 3 1/8 inches and the full-scale shall cover an arc of not less than 90° or more than 150°. The scale shall be legible to normal vision in daylight at a distance of 18 inches.

3.6.2.4.1 Ammeter scale. Unless otherwise specified (see 6.2), the minimum range for the ammeter shall be from minus 5 to 0 to 120-amperes direct current. When specified (see 6.2), additional current ranges using external shunts shall be provided on the ammeter scale. The angle subtended by the smallest scale division shall be not less than 1°. The value of each scale division shall be either one, two, or five of the units measured, or any decimal multiple or submultiple of these numbers. Not more than 100 millivolts shall be required for full scale deflection.

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3.6.2.4.2 Voltmeter scale. Unless otherwise specified (see 6.2), the voltmeter range readings taken at 7.5, 14.5, and 28.5-volts shall be within an accuracy of plus or minus 2 percent of these values. End-scale reading on each range shall be adjusted so that the error in readings taken at the above voltages shall not exceed 1 percent of these voltages when measured on the lowest permissible range. The number of the smallest scale divisions shall be such as to permit readings within the accuracy specified herein by interpolating to not less than one-half of a scale division. The value of each scale division shall be either one, two, or five of units measured, or any decimal multiple or sub-multiple of these numbers. Not more than 10 milliamperes shall be required for end-scale deflection.

3.6.2.5 Zero adjustment. Meters shall be provided with an external zero adjustment which is operable from the front of the meter. The range of the zero adjustment, both above and below the zero point, shall be not less than 3 percent of the full-scale length and shall not loosen when subjected to vibration. The zero adjustment shall be insulated. The adjustment shall be capable of rotating 360° without damage to the instrument or the adjusting screw.

3.6.2.6 Position influence (balance). Position influence, expressed as a percentage of scale length and defined as the maximum displacement of the pointer which is caused solely by a 60° inclination of the meter from normal operating position in the direction which produces the most unfavorable conditions as to position, shall not exceed 2 percent.

3.6.2.7 Damping factor. The damping factor of each meter shall be greater than 2.5. Damping factor is defined as the ratio of the steady deflection to the difference between maximum momentary deflection and steady deflection produced by a sudden application of constant electrical power, of the magnitude the meter was designed to measure. All deflections are measured in angular degrees. For the purposes of this specification, end-scale deflection is used as the maximum momentary deflection.

3.6.2.8 Response time. The length of time required for the pointer of each meter to come to apparent rest, when steady electric power sufficient to produce momentary end-scale deflection is applied, shall not exceed 2 seconds.

3.6.3 Test set circuitry. The test set circuitry shall consist of the external meter circuits with the associated switches, binding posts, fixed resistors, and rheostats.

3.6.3.1 Binding posts. Binding posts shall be made from non-ferrous metal resistant to oxidation. The binding posts shall have high

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electrical conductivity and shall be of a size to carry continuously the maximum current of the circuit involved.

3.6.3.2 Ammeter shunts. When specified (see 6.2), additional shunts shall be required for measuring current beyond the range specified in 3.6.2.4.1. The shunt or shunts, as required, shall be protected by a wrapping of suitable electrical insulation tape or neoprene covering.

3.6.3.3 Load rheostat. A carbon pile rheostat shall be furnished within the test set enclosure and shall be available to use individually or in series with the ammeter. The rheostat shall provide a resistive load infinitely variable over the range required to test 6, 12, and 24-volt current regulators. The heat accumulating from the load rheostat in 48 hours of full-load operation at a duty cycle of one minute "on" and 29 minutes "off" shall not cause injury to the components of the test set or impair its accuracy and the panel temperature, including the case, shall not exceed 40°F. above an ambient temperature of 80° ± 10°F. An automatic overload device shall be provided in the load rheostat circuit.

3.6.3.4 Field rheostat. A tapered wire-wound 0-50 ohm, 50 watt field rheostat shall be inclosed in the test set and shall be provided with overcurrent protection. The rheostat control shall be placed on the instrument panel and shall be plainly marked to show "increase" and "decrease". Binding posts or jacks shall be provided on the panel for field connections.

3.6.3.5 Fixed resistors. Fixed value wire-wound load resistors for 6-volt, 12-volt, and 24-volt systems shall be provided as required and shall be conveniently available through a selector switch or terminals. They shall be fully protected against overloading.

3.6.3.6 Voltage range selector. Unless otherwise specified (see 6.2), range selection shall be made with a rotary selector switch mounted on the test panel. Switch contacts shall be silver plated. Wire-wound, fixed precision resistors shall be used for each range.

3.6.4 Test leads. Test leads shall be provided and shall be single conductor, extra-flexible stranded wire with oil-resistant insulation. The leads shall be of adequate size to withstand the maximum loads required and the normal physical abuse. Ammeter leads shall be not less than 5 feet long, and voltmeter leads not less than 10 feet in length. Insulation on the positive lead shall be of a red color, while the negative lead shall be black. All other leads (in addition to the ammeter and voltmeter leads), clips, and connectors required for the intended use of the test set shall be provided.

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3.7 Electrical components. Electrical components of the test set, including switches and wiring, shall conform to and shall be located in accordance with the NEMA Standards, except as specified herein. No component part shall be used in any application where the requirements exceed the manufacturer's rating for the part.

3.7.1 Wiring. The internal wiring of the test sets shall be color coded in accordance with the requirements of MIL-STD-681.

3.8 Fungus control. Unless otherwise specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with type II requirements of MIL-T-152, except that the statement of treatment is not required.

3.9 Painting. Unless otherwise specified (see 6.2), painting and finishing of the test set shall be in accordance with the supplier's standard practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel shall have been applied.

3.10 Identification of product. The test set shall be marked in accordance with MIL-STD-130 and when specified (see 6.2) shall include the Federal Stock Number (FSN). Unless otherwise specified (see 6.2), identification marking shall be accomplished on a corrosion-resistant metal nameplate securely attached to the test set.

3.11 Handbook of instructions. Unless otherwise specified (see 6.2), the contractor shall furnish two copies of a handbook of instructions sealed in a waterproof envelope securely attached to each test set supplied under this specification. The handbook shall contain necessary operating instructions, drawings, illustrations, wiring diagrams, parts list, and necessary periodic adjustments to insure proper operation of the test set. Additional copies of the instruction handbook and the mailing address to which additional copies shall be sent shall be as specified by the procuring activity (see 6.2).

3.11.1 Handbook approval. When specified (see 6.2), a preliminary copy of the instruction handbook shall be submitted to the contracting officer for approval prior to publication and delivery.

3.12 Workmanship. Workmanship of the test set shall be of the acceptable quality prevailing among manufacturers normally producing test sets of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).

(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection applies to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) as follows:

(a) Examination (4.3).

(b) Initial accuracy test (4.4.1).

(c) Inspection of preparation for delivery (4.5).

4.2.2.1 Sampling. When specified (see 6.2), sampling for lot acceptance shall be in accordance with MIL-STD-105 at the inspection level and acceptable quality level (AQL) specified in the contract. Samples for the test shall be selected at random from the samples that passed the examination in 4.3.

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4.3 Examination. Each production item or sample shall be visually examined to determine compliance with requirements in section 3, using the following check list as a guide:

Examination check list	Requirement reference
Design features not as specified	3.2
Construction not as specified	3.4 through 3.4.9
Dimensions not as specified	3.5.10
Enclosure not as specified	3.6.1
Instrument panel not as specified	3.6.1.1
Meters not as specified	3.6.2 through 3.6.2.4
Zero adjustment not as specified	3.6.2.5
Position influence (balance) not as specified	3.6.2.6
Test leads not as specified	3.6.4
Fungus control not as specified	3.8
Identification of product not as specified	3.10
Handbook of instructions omitted or incomplete	3.11
Workmanship not as specified	3.12

4.4 Test methods.

4.4.1 Accuracy tests. Accuracy of the meter circuits shall be measured on each range for ascending and descending readings at not less than 6 points equally spaced over the entire scale. Readings shall be taken with the test set in its normal operating position and with the test leads and external shunts, when provided, attached. (Tapping of meters is prohibited.) With the exception of 4.4.3, tests shall be conducted in an ambient air temperature of 80° ± 10°F. Accuracy of measurement shall be determined by use of standard meters having an accuracy range of within one-fourth of one percent.

4.4.1.1 Initial accuracy. The initial accuracy of the test set shall be within 2 percent of end-scale value for the voltmeter and within 3 percent of full-scale value for the ammeter. Failure of the test set to operate within these accuracies shall be cause for rejection of the test set (see 3.5.1).

4.4.1.2 Accuracy after tests. Accuracy tests following the tests specified in 4.4.2, 4.4.4, 4.4.6, 4.4.7, 4.4.8, 4.4.9, 4.4.10, and 4.4.12 shall show accuracies within 1 percent of full-scale value from the readings taken as specified in 4.4.1.1. Failure of the test set to operate within these accuracies shall be cause for rejection of the test set (see 3.5.2).

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4.4.2 Storage tests.

4.4.2.1 Low temperature test. The test set and all components thereof, shall be placed in an ambient-air temperature of minus 60°F. to -65°F. for a period of 72 hours. At the end of this time, the test set shall be removed and allowed to stabilize at an air temperature of plus 80° ± 10°F. The test set shall then be inspected for evidence of damage and tested for accuracy in accordance with 4.4.1.2 (see 3.5.3.1).

4.4.2.2 High temperature test. The test set and all components thereof shall be submitted to a 24-hour cycle of temperature changes starting with 10 hours at plus 90° ± 5°F., followed by 5 hours of steady increase to plus 150° to + 155°F., 4 hours at plus 150° to + 155°F., and 5 hours of steady decrease to plus 90° ± 5°F. At the end of this time the test set shall be removed and allowed to stabilize at an air temperature of plus 80° ± 10°F. The test set shall then be inspected for evidence of damage and tested for accuracy in accordance with 4.4.1.2 (see 3.5.3.1).

4.4.3 Operating temperature tests.

4.4.3.1 Low-ambient temperature test. The test set after being in an ambient-air temperature of minus 20° to -25°F. until stabilized, and while at that temperature, shall be tested for accuracy in accordance with 4.4.1. Accuracy readings taken at this temperature shall not vary more than 2 percent of full-scale value from the readings taken as specified in 4.4.1.1 (see 3.5.3.2.1).

4.4.3.2 High-ambient temperature test. The test set after being in an ambient-air temperature of plus 110° to + 115°F. until stabilized, and while at that temperature, shall be tested for accuracy in accordance with 4.4.1. Accuracy readings taken at this temperature shall not vary more than 1 percent of full-scale value from the readings taken as specified in 4.4.1.1 (see 3.5.3.2.2).

4.4.4 Humidity test. The test set and all components thereof shall be subjected to 20 hours exposure with relative humidity between 93 and 97 percent and at air temperatures from plus 80° to + 85°F., followed by 4 hours of 100 percent relative humidity with condensation at plus 75° to + 80°F.; air movement and radiation negligible. The test set shall then be inspected for evidence of damage and tested for accuracy in accordance with 4.4.1.2 (see 3.5.3.3).

4.4.5 Meter characteristic tests. Characteristic tests for the meters shall be conducted in an ambient-air temperature of 80° ± 10°F. with the meters in their normal operating position. Failure of the meters to meet any of the characteristics tests specified herein shall be cause for rejection of the test set.

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4.4.5.1 Full and end scale deflection tests. A series of five readings shall be taken to determine the average voltage required to produce full-scale deflection of the ammeter. This value shall not exceed 100 millivolts (see 3.6.2.4.1). Five readings shall also be taken to determine the average current required to produce end-scale deflection of the voltmeter. This value shall not exceed 10 milliamperes (see 3.6.2.4.2).

4.4.5.2 Zero adjustment test. The range of the zero adjustment shall be determined by test. There shall be a minimum adjustment of 3 percent of the full-scale length available both above and below the zero mark. The zero adjuster shall be rotated 360° in both directions. This rotation shall not cause damage to the meter or the adjusting screw (see 3.6.2.5).

4.4.5.3 Position influence test. The pointer shall be set in the center of the zero-scale division. The meter shall then be tilted 60° from this position in the direction producing the most unfavorable conditions as to position. The deviation shall not exceed 2 percent of the scale length (see 3.6.2.6).

4.4.5.4 Damping factor test. Voltage or current, as required for the voltmeter and ammeter, respectively, sufficient to produce momentary end-scale deflection shall be observed and recorded. After the pointer has come to rest, the steady deflection value shall be recorded. The damping factor shall be the ratio of the steady deflection (in angular degrees) to the difference between the maximum deflection and the steady deflection (in angular degrees). The damping factor shall be not less than 2.5 (see 3.6.2.7).

4.4.5.5 Response time test. Voltage or current, as required for the voltmeter and ammeter, respectively, sufficient to produce momentary end-scale deflection, shall be applied to the meter circuits. The length of time in seconds required for the pointer to come to apparent rest shall be recorded. The pointer shall be considered as having come to apparent rest when it has attained the actual point of rest within 1 percent of full-scale value. The test shall be repeated at least four times, and the average length of time recorded for all the measurements shall be considered as the response time of the meter. Response time shall not exceed 2 seconds (see 3.6.2.8).

4.4.6 Momentary overload test. The meters of the test set shall be subjected to a series of momentary overloads as specified herein. Upon completion, except for setting of the pointer to zero using the external adjustment screw, no additional repairs or adjustments shall be required. Failure of the test set to meet the accuracy tests of 4.4.1.2 shall be cause for rejection (see 3.5.4).

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4.4.6.1 Ammeter circuit. The ammeter, with any one of its shunts, shall be subjected to momentary overloads three times full-scale value for a period of 1 second "on" and 60 seconds "off" for a total of five cycles, followed by a 5 second "on" period (see 3.5.4.1).

4.4.6.2 Voltmeter circuit. The voltmeter shall be subjected to momentary overloads of 2.8 times end-scale value for periods specified in 4.4.6.1 (see 3.5.4.2).

4.4.7 Dielectric strength test. The test set shall be subjected to 500 volts r.m.s. (60 hertz approximate sine wave form) applied for 1 minute between mutually insulated circuits and between insulated circuits and external metal parts. The test shall be conducted following the momentary overload test of 4.4.6. The test set shall then be tested for accuracy in accordance with 4.4.1.2 and inspected for evidence of damage. Failure of the test set to meet the accuracies specified or evidence of damage shall be cause for rejection of the test set (see 3.5.8).

4.4.8 Sustained overload test. The meters of the test set shall be subjected to an overload of 20 percent greater than end-scale values, sustained for a period of 8 hours. The test set shall then be tested for accuracy in accordance with 4.4.1.2. Failure of the test set to meet the accuracies specified or evidence of damage shall be cause for rejection of the test set (see 3.5.5).

4.4.9 Vibration test. The test set shall be subjected, along each of its three major axes, to a vibration (simple harmonic motion) having an amplitude of 1/32 inch (1/16 inch maximum total excursion). The frequency shall be varied uniformly between the limits of 5 and 55 c.p.s. and back to 5 c.p.s., 1 traverse per minute, for a period of 1 hour. The test set shall then be tested for accuracy in accordance with 4.4.1.2. Failure of the test set to meet the accuracies specified or evidence of damage shall be cause for rejection of the test set (see 3.5.6).

4.4.10 Shock test. The test set shall be dropped a distance of 8 inches, along each of its three major axes, onto a concrete surface. The test set shall then be tested for accuracy in accordance with 4.4.1.2. Failure of the test set to meet the accuracies specified or evidence of damage shall be cause for rejection of the test set (see 3.5.7).

4.4.11 Contact life test. Rheostat contacts and all switches furnished shall be cycled continuously through all contacts at a rate of 10 to 16 c.p.m. for a total of 5,000 cycles each. The switch shall carry a current of 150 percent of the maximum normal current in its

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electrical circuit. Before and after this test, the voltage drop across any switch or contact shall not exceed 50 millivolts when the switch or contact is in the closed position. The rheostat shall not show an open circuit at any position other than at the "open" position. Failure of the test set to meet the tests specified herein shall be cause for rejection (see 3.5.9).

4.4.12 Overheating test. The test set shall be operated at maximum rated load for a minimum period of 48 hours. The rheostat load switch during this time shall operate for one minute "on" and 29 minutes "off". During and following this test, neither the panel or the case temperature shall exceed the ambient temperature (80° ± 10°F.) by more than 40°F. The test set shall then be tested for accuracy in accordance with 4.4.1.2. Evidence of damage to components or failure to meet the accuracies specified shall be cause for rejection (see 3.6.3.3).

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the test set shall be inspected to determine conformance with the applicable requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation and packaging shall be for level A or C and packing shall be for level A, B, or C of MIL-E-17555 as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test set covered by this specification is intended for "on the vehicle" testing of automotive low-voltage, non-waterproof electrical systems (6, 12, and 24 volts), incorporated in passenger and commercial type vehicles. Test sets for testing military tactical vehicles using waterproof electrical systems should be purchased under MIL-T-10308.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) First article, when required (see 3.1).

(c) Special safety devices, when required (see 3.4.6).

(d) Case dimensions, if different (see 3.5.10).

(e) Windows, if different (see 3.6.2.3).

(f) Ammeter and voltmeter ranges, if different (see 3.6.2.4.1 and 3.6.2.4.2).

(g) Additional current ranges, when required (see 3.6.2.4.1).

(h) Additional shunts, when required (see 3.6.3.2).

(i) Range selection switch, if different (see 3.6.3.6).

(j) Fungus control, if different (see 3.8).

(k) Painting, if different (see 3.9).

(l) FSN when required and identification marking, if different (see 3.10).

(m) Handbook of instructions, if different, and address where additional copies are to be sent (see 3.11).

(n) Preliminary Handbook approval, when required (see 3.11.1).

(o) First article inspection, if different (see 4.2.1).

(p) Quality conformance inspection, if different (see 4.2.2).

(q) Sampling for lot acceptance, when required (see 4.2.2.1).

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(r) Selection of applicable level of packaging and packing required (see 5.1).

MILITARY INTEREST:

Custodians:
Army - WC
Navy - MC
Air Force - 82

Review activities:
Army - EL, SM
Navy - MC
Air Force - 82

User activities:
Army - AT, ME
Navy - SH

Preparing activity:
Army - WC

CIVIL AGENCIES INTEREST:
GSA - FSS
VA - DMS
HEW

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price ? cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

W-T-215C TEST SET; GENERATOR AND VOLTAGE REGULATOR, AUTOMOTIVE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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